

SIMERA

SENSE

xScape100 (with Enclosure)

Interface Control Document

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List of Abbreviations

Abbreviation	Description
AIT	Assembly, Integration and Testing
BOL	Beginning of Life
CE	Control Electronics
CVCM	Collected Volatile Condensable Material
DC	Direct Current
dTDI	Time Delayed Integration in the Digital Domain
FEE	Front-End Electronics
FFOV	Full Field of View
FWHM	Full Width at Half Maximum
HPP	Half Power Point
HS	Hyperspectral
IO	Input/Output
ISO	International Organization for Standardization
LVC MOS	Low Voltage Complementary Metal Oxide Semiconductor
LVDS	Low Voltage Differential Signalling
MIT	Mounting Interface Temperature
MS	Multispectral
NC	Not Connected
NIR	Near-Infrared
OFE	Optical Front-End
PAN	Panchromatic
PCB	Printed Circuit Board
SPI	Serial Peripheral Interface
TBC	To Be Confirmed
TML	Total Mass Loss
VIS	Visible
VNIR	Visible and Near-Infrared

1. Introduction

1.1 Identification

Item Description: Imager products with electronics enclosures in the xScape100 Product Range, as listed in Table 1-1, hereafter referred to as the “Imager”.

Table 1-1: Product Matrix

		Product Range
		xScape100
Product Type	MonoScape	MonoScape100 (with Enclosure) 055410
	TriScape	TriScape100 (with Enclosure) 055411
	MultiScape CIS	MultiScape100 CIS (with Enclosure) 055412
	HyperScape	HyperScape100 (with Enclosure) 055413

1.2 Intended Use

This document describes the interfaces and environmental operating conditions of Imagers with electronics enclosures in the xScape100 Product Range.

1.3 Context and Summary

The xScape100 (with Enclosure) Product Range is produced by Simera Sense and includes electro-optical snapshot and push-broom imaging systems (Imagers). The Imagers in this product range are intended for earth observation applications and are primarily designed to be implemented as part of an optical payload in a nanosatellite structure; however, it can also be used as part of larger satellite systems.

Imagers in this product range differ from the Imagers in the xScape100 Product Range in that they offer a mechanical enclosure which fully encloses all Imager Electronics. This enclosure offers: additional ingress protection, increased radiation shielding and enables compliance to the requirement that all components of the payload must be fully enclosed, as is stipulated by a number of satellite bus providers.

The Imager is any of the products as listed in Table 1-1:

- The MonoScape100 (with Enclosure) is a snapshot imaging system employing a monochrome CMV12000 sensor. The Imager is not fitted with a NIR blocking filter.
- The TriScape100 (with Enclosure) is a snapshot imaging system employing a colour sensor with integrated RGB Bayer filter having 'RGGB' order. It is fitted with a NIR blocking filter.
- The MultiScape100 CIS (with Enclosure) is a push-broom imaging system providing a panchromatic plus 7 multispectral bands in the Visible and Near Infrared (VNIR) spectrum.
- The HyperScape100 (with Enclosure) is a hyperspectral imaging system providing a panchromatic plus over 400 hyperspectral bands in the VNIR spectrum.

This Interface Control Document identifies, defines, and describes the interfaces between the Imager and the surrounding satellite components, as well as between the Imager and its environment. Information presented in this document is applicable to the Imagers as identified in Table 1-1.

2. Referenced Documents

Table 2-1 lists documents that are referenced in this document. In the event of conflict between the contents of the reference documents and this document, this document shall take precedence.

Table 2-1: Referenced Documents

Ref. #	Reference
[1]	Outgassing Data for Selecting Spacecraft Materials. [Online]. Available: https://outgassing.nasa.gov/ [2018, October 24]
[2]	051715 – xScape Control and Data Interface Control Document
[3]	050443 – xScape SpaceWire Interface Control Document
[4]	051967 – xScape RS422,485 Interface Control Document
[5]	052024 – xScape USART Interface Control Document
[6]	053036 – xScape Data Encryption Interface Control Document
[7]	055324 – xScape Image Data Compression Interface Control Document
[8]	078410 – xScape100 (with Enclosure) Reduced Thermal Model Document

For undated references, the latest released version of the reference document applies. For dated references, subsequent versions of the document do not apply. It is best practice to always refer to the latest released version. Unless otherwise stated, web links referenced above were last accessed at the release date of the current version of this document.

3. System Description and Context

The Imager captures electromagnetic radiation, focuses the radiation on a sensor and converts the focused electromagnetic radiation into electrical signals. The Imager typically forms part of the payload of a satellite and is shown in the context of a typical system diagram in Figure 3-1.

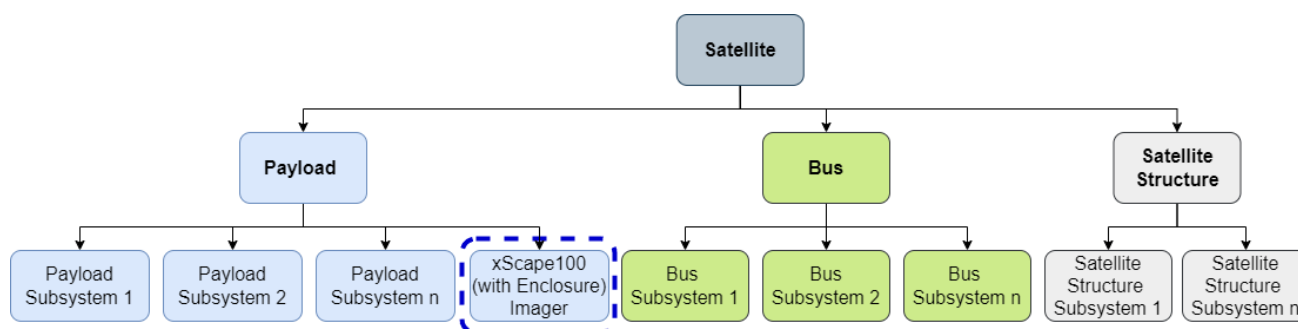


Figure 3-1: Typical Satellite System Diagram

3.1 Physical Description

The Imager consists of several subassemblies which comprises of the xScape100 Optical Front-End (OFE), the xScape100 Sensor Unit (with Enclosure), the xScape100 Control Electronics and the xScape100 Electronics Enclosure. Table 3-1 provides a functional description of the Imager and its components.

Table 3-1: System and Component Functional Description

ID	Imager/Component	Primary Function
1	Imager	Collects electromagnetic radiation and converts it into electrical signals
1.1	xScape100 Optical Front-End	Focuses the collected electromagnetic radiation onto an imaging sensor
1.2	xScape100 Sensor Unit (with Enclosure)	Positions the sensor on the focal plane and converts the focused electromagnetic radiation into electrical signals
1.3	xScape100 Control Electronics	Powers and drives the sensor unit, as well as provides storage space for the captured images
1.4	xScape100 Electronics Enclosure	Encloses and shields the Sensor Unit and Control Electronics

Figure 3-2 illustrates the physical system subassemblies and provides the axis definition. The xScape100 Electronics Enclosure encloses both the xScape100 Sensor Unit (with Enclosure) and the xScape100 Control Electronics. An exploded view of the aforementioned is shown in Figure 3-3, where the xScape100 Electronics Enclosure has been excluded.

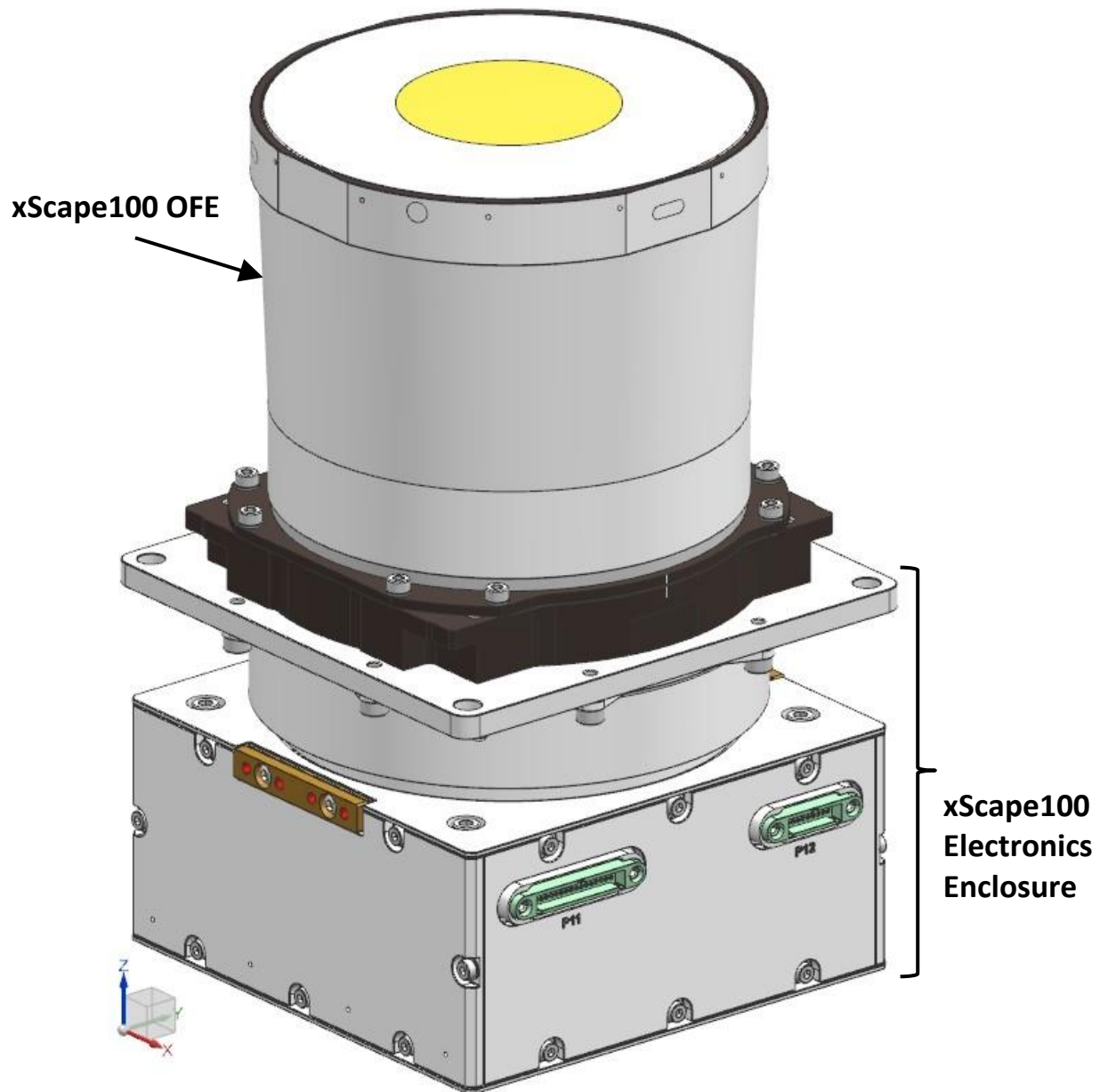


Figure 3-2: Imager with Axis Definition

An exploded view of the xScape100 Sensor Unit (with Enclosure) and the xScape100 Control Electronics is shown in Figure 3-3. Refer to 4.5.1.1 for the physical electrical interface options.

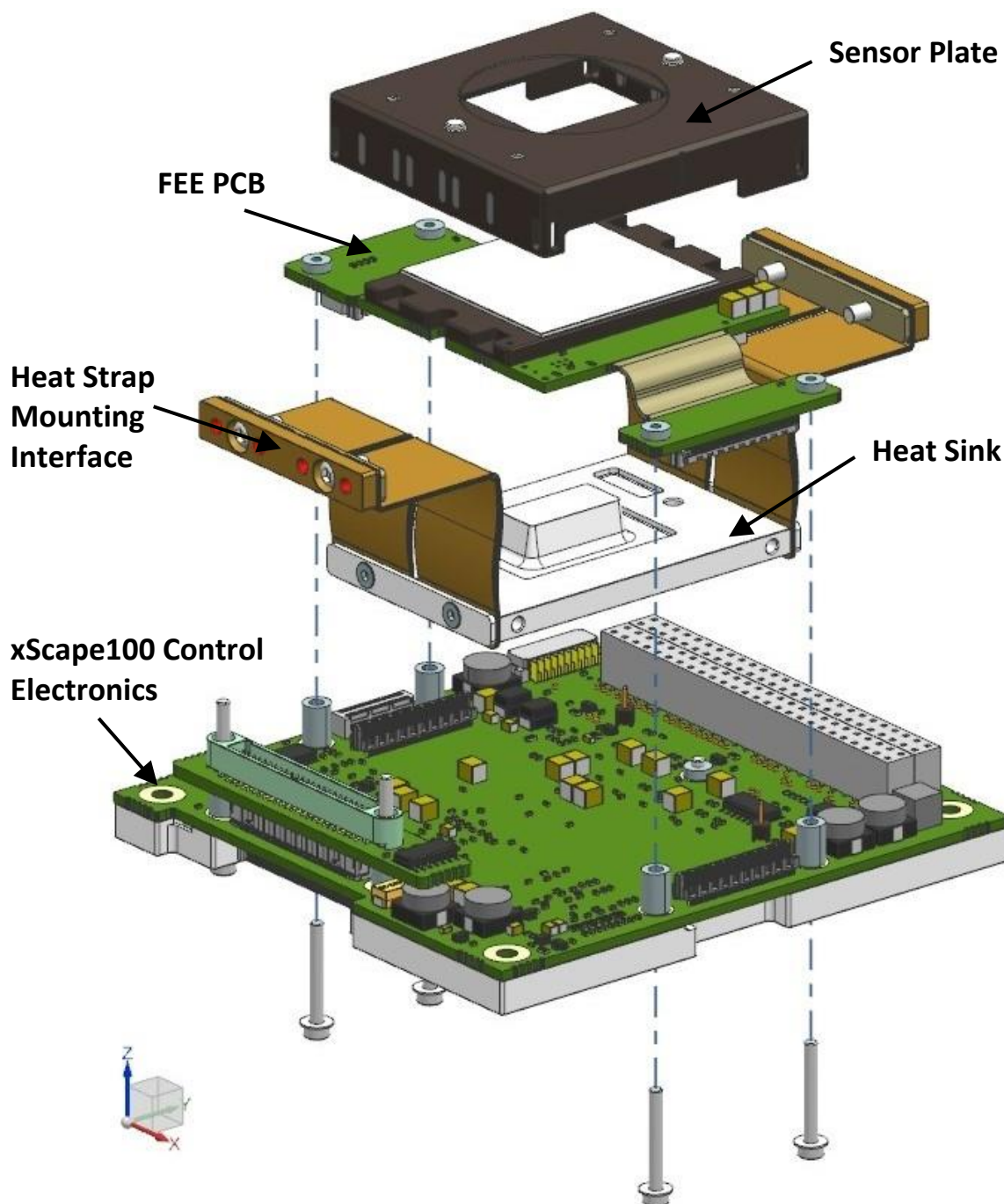


Figure 3-3: Exploded View of xScape100 Sensor Unit (with Enclosure) and Control Electronics

3.2 Physical Properties

The physical properties of the Imager are presented in Table 3-2. The reference axis system used to define the centre of mass position is shown in Figure 3-4. The origin of the reference axis system used to define the moments of inertia and centre of mass position is located on the OFE Mounting Interface, at the geometric centre of the OFE's mounting points (which are shown in Figure 4-5).

Table 3-2: Physical Properties

Property	Unit	Value
Mass	kg	1.58 (± 5%)
Centre of Mass Position		
X	mm	0.13 (± 0.5) See Figure 3-4
Y	mm	-0.76 (± 0.5) See Figure 3-4
Z	mm	-5.4 (± 1) See Figure 3-4

Equation 1 provides the moments of inertia (I) relative to the reference axis system shown in Figure 3-4. The origin of the reference axis system used to define the moments of inertia, is located on the OFE Mounting Interface, at the geometric centre of the OFE's mounting points. Equation 2 provides the moments of inertia (I_C) about the Imager's centre of mass. The position of the centre of mass is shown in Figure 3-4. Equation 3 provides the principal moments of inertia (I_P). All moments of inertia values are presented in units of kg.m^2 .

$$I [\text{kg.m}^2] = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} = \begin{bmatrix} 5.4 \times 10^{-3} & 5.4 \times 10^{-6} & -7.8 \times 10^{-6} \\ 5.4 \times 10^{-6} & 5.3 \times 10^{-3} & 79.1 \times 10^{-6} \\ -7.8 \times 10^{-6} & 79.1 \times 10^{-6} & 2.8 \times 10^{-3} \end{bmatrix} \quad (1)$$

$$I_C [\text{kg.m}^2] = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}_C = \begin{bmatrix} 5.4 \times 10^{-3} & 5.5 \times 10^{-6} & -6.6 \times 10^{-6} \\ 5.5 \times 10^{-6} & 5.3 \times 10^{-3} & 72.6 \times 10^{-6} \\ -6.6 \times 10^{-6} & 72.6 \times 10^{-6} & 2.8 \times 10^{-3} \end{bmatrix} \quad (2)$$

$$I_P [\text{kg.m}^2] = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} = \begin{bmatrix} 5.4 \times 10^{-3} & 0 & 0 \\ 0 & 5.3 \times 10^{-3} & 0 \\ 0 & 0 & 2.8 \times 10^{-3} \end{bmatrix} \quad (3)$$

Figure 3-4 presents the envelope dimensions of the Imager and indicates the position of the centre of mass relative to the OFE's mounting points.

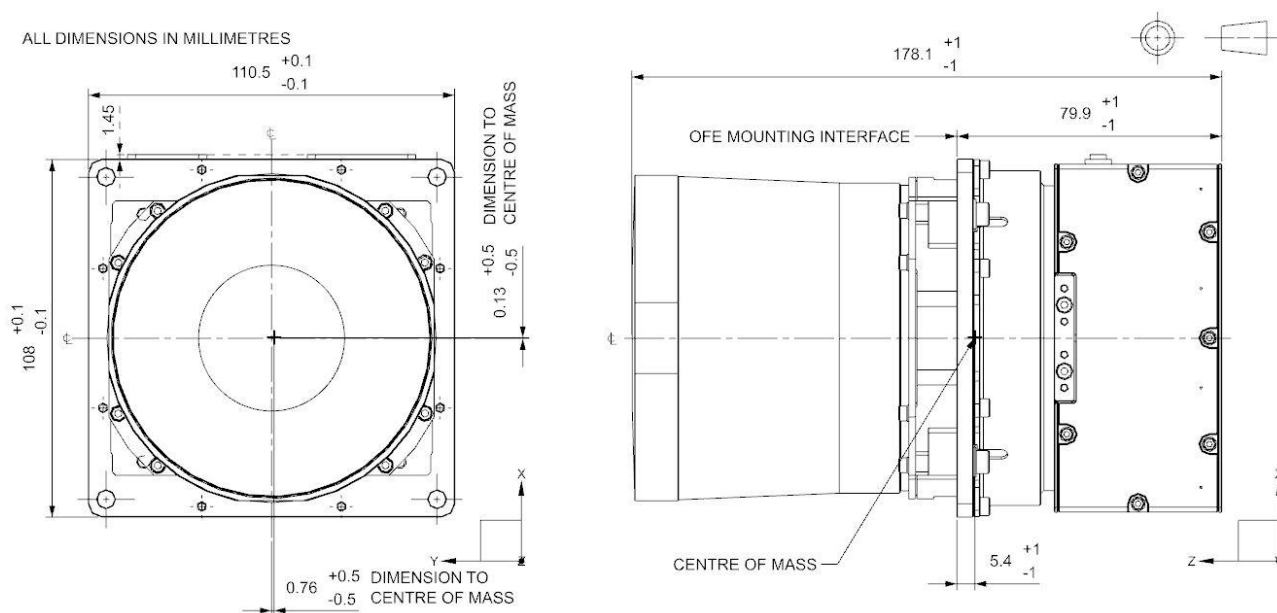
**Figure 3-4: Centre of Mass Position**

Table 3-3 provides the natural frequencies of the Imager (between 0 Hz and 2000 Hz) when it is constrained at its four mounting points, which are shown in Figure 4-5. The values presented in Table 3-3 were obtained through finite-element analysis with the Imager being mounted to the vibration jig (material Aluminium 6082-T6). Note: The structural behaviour of the Imager is influenced by the mechanics of the satellite bus structure. The modes presented in Table 3-3 are therefore dependant on the mechanics to which the Imager is mounted. The modes in Table 3-3 will differ from those presented here if the satellite mounting mechanics have a different stiffness from that of the vibration jig.

Table 3-3: Natural Frequencies of the Imager

Mode Number	Frequency [Hz]
1	620
2	657
3	914
4	1018
5	1124
6	1245
7	1260
8	1284
9	1579
10	1620
11	1671
12	1790
13	1884

4. Description of System Interfaces

4.1 Interface Identification and Definition

The various interfaces between the Imager and the satellite components, as well as between the Imager and its environment, are shown graphically in Figure 4-1 below. The satellite components are herein defined as being all components which do not form part of the Imager and as such include the satellite structure.

Note: For analysis purposes it is assumed that there is no thermal interface via the wiring harnesses.

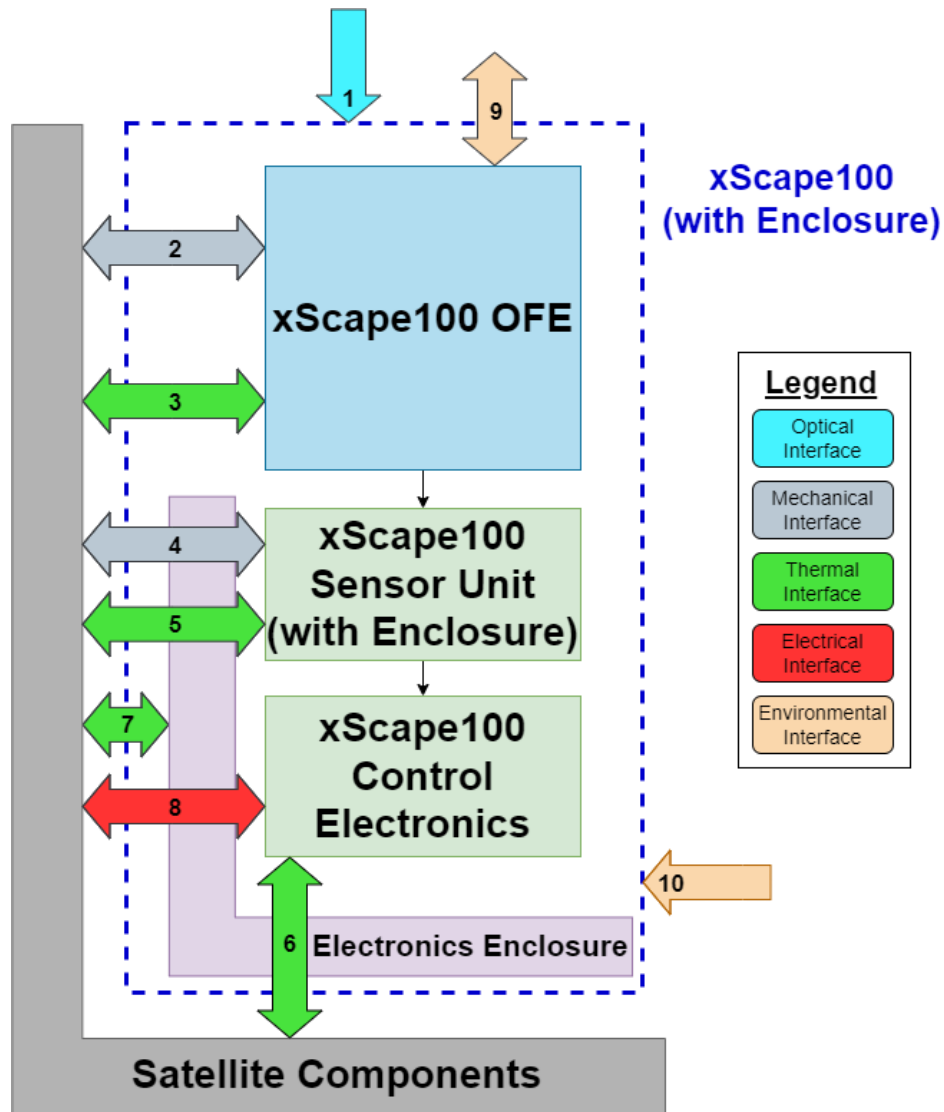


Figure 4-1: Interface Identification Diagram

The interfaces identified in Figure 4-1, are defined in Table 4-1 below. The descriptions of the interfaces are presented in the subsections following Figure 4-1.

Table 4-1: Interface Definition

Interface Number	Interface Type	Interface From	Interface To
1	Optical	Target in View	xScape100 (with Enclosure)
2	Mechanical	xScape100 OFE	Satellite Components
3	Thermal	xScape100 OFE	Satellite Components
4	Mechanical	xScape100 Sensor Unit (with Enclosure)	Satellite Components
5	Thermal	xScape100 Sensor Unit (with Enclosure)	Satellite Components
6	Thermal	xScape100 Control Electronics	Satellite Components
7	Thermal	xScape100 Electronics Enclosure	Satellite Components
8	Electrical (Power and Control)	Satellite Components	xScape100 Control Electronics
9	Environmental (Thermal Radiation)	xScape100 OFE	Environment
10	Environmental (Cosmic Radiation)	Environment	xScape100 (with Enclosure)

4.2 Optical Interface

4.2.1 Interface 1: Target in View to xScape100 (with Enclosure)

The Imager has an optical interface at its front aperture with a diameter of 95 mm and full field of view (FFOV) of 2.96°. The Imager's *effective* across-track FFOV is 2.22°. The *effective* along-track FFOV is 1.67° for the snapshot Imagers (MonoScape100 and TriScape100) and although the same sensor is used across the product range, the *effective* along-track FFOV is less for the line scan Imagers (MultiScape100 CIS and HyperScape100) since the entire sensor active area is not utilised.

The function of this optical interface is to enable the collection of electromagnetic radiation by the Imager. This optical interface of 95 mm diameter with FFOV of 2.96° shall remain unobscured during imaging to ensure optimal performance of the Imager.

Figure 4-2 shows the Imager ground projection for snapshot Imagers including the position of the first pixel, PIXEL (0,*n*), clocked out from each image line *n* (starting with line 0) with respect to the Imager axis definition. The Bayer colour filter array order is also shown, applicable only to the TriScape100.

Figure 4-3 shows the Imager ground projection for line scan Imagers including the position of the first pixel, PIXEL (0), clocked out from each line in each band with respect to the Imager axis definition. Band 0 is reserved for the panchromatic (PAN) band. The multispectral (MS) bands are assigned in increasing wavelengths, e.g., the centre wavelength of MS 3 will be lower than that of MS 4. The scan direction determines which band is read out first: image lines from each enabled band is stored into the session as they scan over the target.

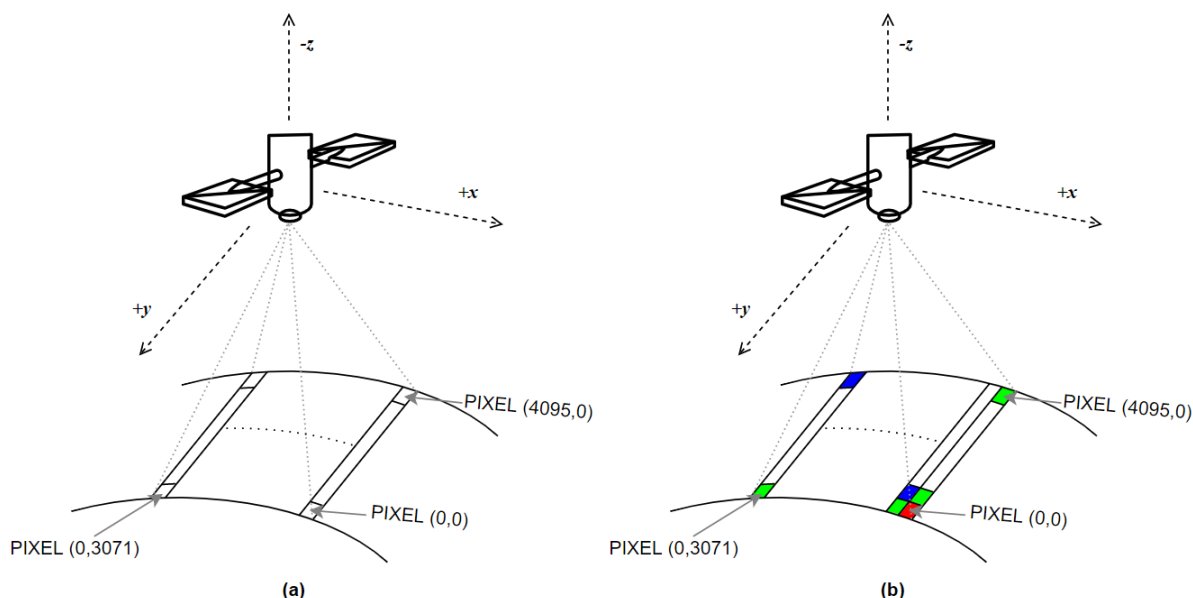


Figure 4-2: Imager Ground Projection for (a) MonoScape100, and (b) TriScape100

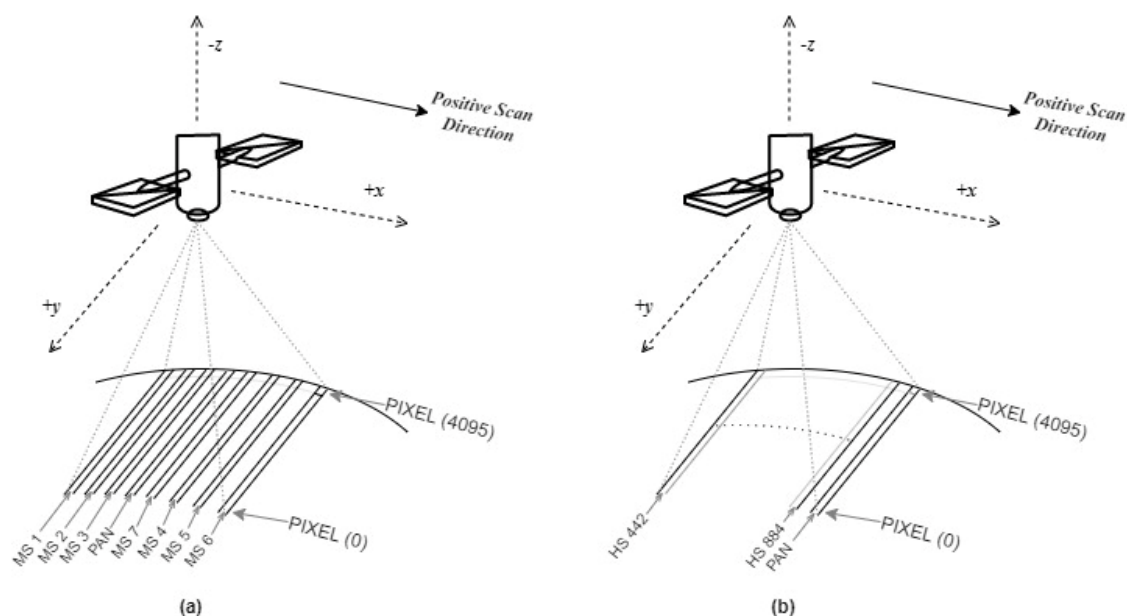


Figure 4-3: Imager Ground Projection for (a) MultiScape100 CIS, and (b) HyperScape100

The maximum sun exposure time (the maximum duration the Imager may continuously have the sun directly within its field of view with the sensor not powered) shall be limited to 5 minutes. The maximum sun exposure time is specified at 5 minutes to avoid the sensor exceeding its maximum rated temperature.

Figure 4-4 shows the point source transmittance of the Imager using the stray light baffle design of the system. Figure 4-4 reports the power on the sensor (shown on the y-axis) when observing a source of collimated light (with an output of 1 W) as it is moved from the on-axis position (0°) to 10° off-axis relative to the optical axis of the Imager. Note: the degrees indicated on the x-axis are the half-angles as measured from the optical axis of the Imager.

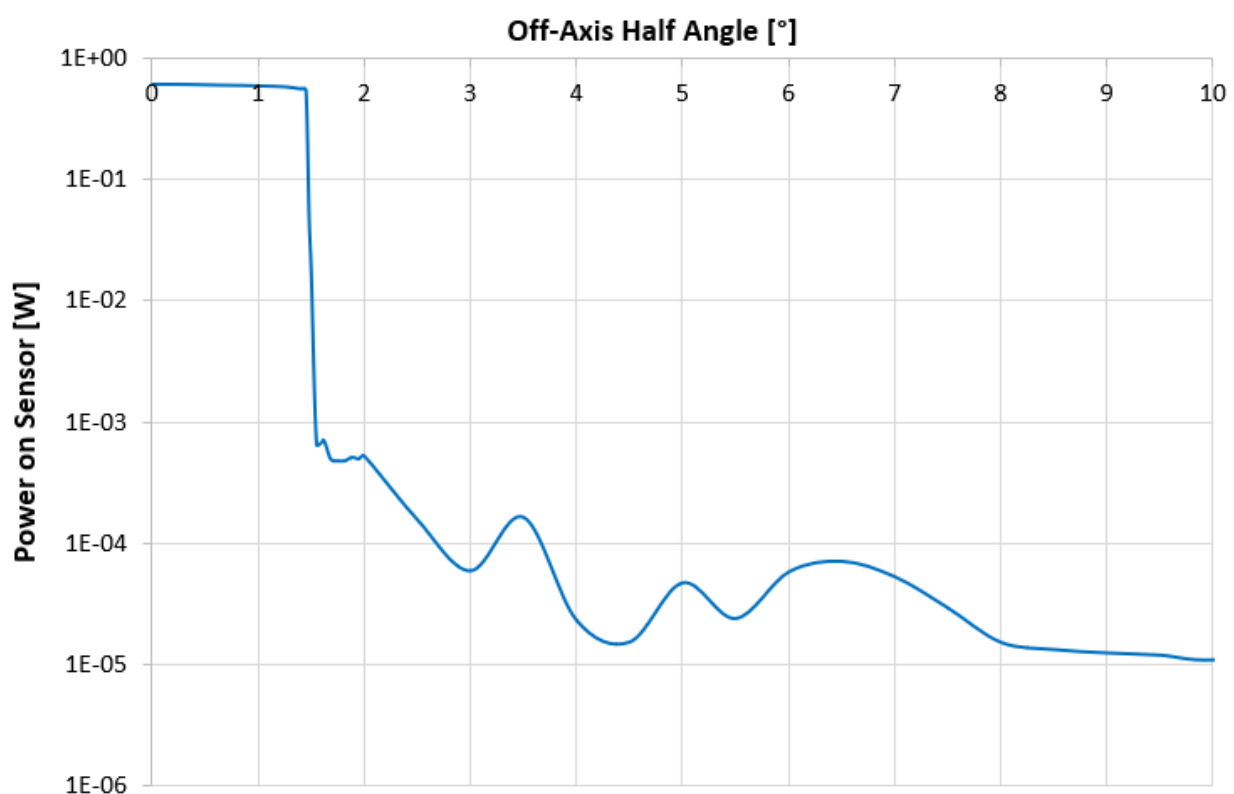


Figure 4-4: Point Source Transmittance of Imager

Based on the data presented in Figure 4-4, it is recommended that a sun exclusion half-angle of 3.75° is used to ensure satisfactory stray light suppression. A sun exclusion half-angle of 3.75° results in less than $1/10000$ of the power entering the Imager from outside the field of view of the Imager reaching the sensor. For angles larger than 10° , the transmittance continues to decrease even further and never rises above 1×10^{-4} of the input power again.

The electromagnetic radiation collected from the target is filtered by the OFE as well as the spectral filter, as and when fitted. For snapshot Imagers, the platform shall track and point the Imager to the target. For line scan Imagers, the target is scanned in the direction of the Imager x-axis, in push-broom imaging fashion. The scanning speed shall therefore stay constant for the duration of imaging. The Imager shall be nadir pointing and roll and yaw shall be kept stable for the duration of imaging. Platform jitter shall be minimized in all cases for optimal image quality.

The TriScape100 Imager is fitted with the xScape100 VIS OFE to suppress the near-infrared spectrum to enable colour imagery to be captured. The rest of the Imagers are fitted with the xScape100 VNIR OFE. Snapshot Imagers can take bursts of up to 128 images at up to 120 frames per second, while line scan Imagers are able to capture images with programmable line rate, strip length and band combinations.

Table 4-2 summarises the optical interface and filter specifications of the Imagers.

Table 4-2: Optical Interface Specification Matrix

Imager	Acquisition	OFE	Filter Specification					
MonoScape100	Snapshot	xScape100 VNIR	-					
TriScape100	Snapshot	xScape100 VIS	RGB Bayer in RGGB order					
MultiScape100 CIS	Line Scan	xScape100 VNIR	Band	Centre Wavelength (nm)	FWHM Bandwidth (nm)	HPP Cut-On (nm)	HPP Cut-Off (nm)	Approximate Sensor Line Number
			#0: PAN	625	250	500.0	750.0	1536
			#1: MS 1	490	65	457.5	522.5	2176
			#2: MS 2	560	35	542.5	577.5	1964
			#3: MS 3	665	30	650.0	680.0	1748
			#4: MS 4	705	15	697.5	712.5	1108
			#5: MS 5	740	15	732.5	747.5	896
			#6: MS 6	783	20	773.0	793.0	680
			#7: MS 7	842	115	784.5	899.5	1324
HyperScape100	Line Scan	xScape100 VNIR	Band	Centre Wavelength	FWHM Bandwidth	Approximate Sensor Line Number		
			#0: PAN	625 nm	250 nm	340		
			#1-31: HS <i>n</i>	460 nm – 860 nm in 1 nm increments	~ 3.5% of CWL + 0.25 nm x N _{dTDI}	row 980 at 860 nm through row 2656 at 460 nm		
			<u>Note:</u> The Hyperspectral bands’ FWHM bandwidth depends on the band’s centre wavelength as well as the number of dTDI stages (N _{dTDI}) applied to the band.					

4.3 Mechanical Interfaces

4.3.1 Interface 2: xScale100 OFE to Satellite Components

The OFE interfaces mechanically with the satellite structure via four through holes located in the OFE's Mounting Interface Plate. The function of this interface is to secure the OFE to the applicable satellite component and act as the main structural support for the Imager.

The mechanical interface to which the OFE shall be secured shall have four M5 threaded holes. These holes shall be spaced to match the hole spacing of the four 5.4 mm diameter through holes (as shown in Figure 4-5) exactly and shall have a positional tolerance of 0.1 mm. All interfaces which mate directly to the OFE mounting interface shall be coplanar and have a flatness tolerance of 0.05 mm or smaller. In addition, the mating interfaces shall have a N7 or smoother surface finish (this is equivalent to a surface finish with an average roughness of $R_a = 1.6 \mu\text{m}$).

Caution shall be exercised during the assembly of any mechanics to the mounting interface of the OFE. All mechanics which mate to the OFE Mounting Interface shall induce a total flatness error of less than 0.05 mm to the OFE Mounting Interface. Figure 4-5 provides the dimensions of the OFE's mounting interface.

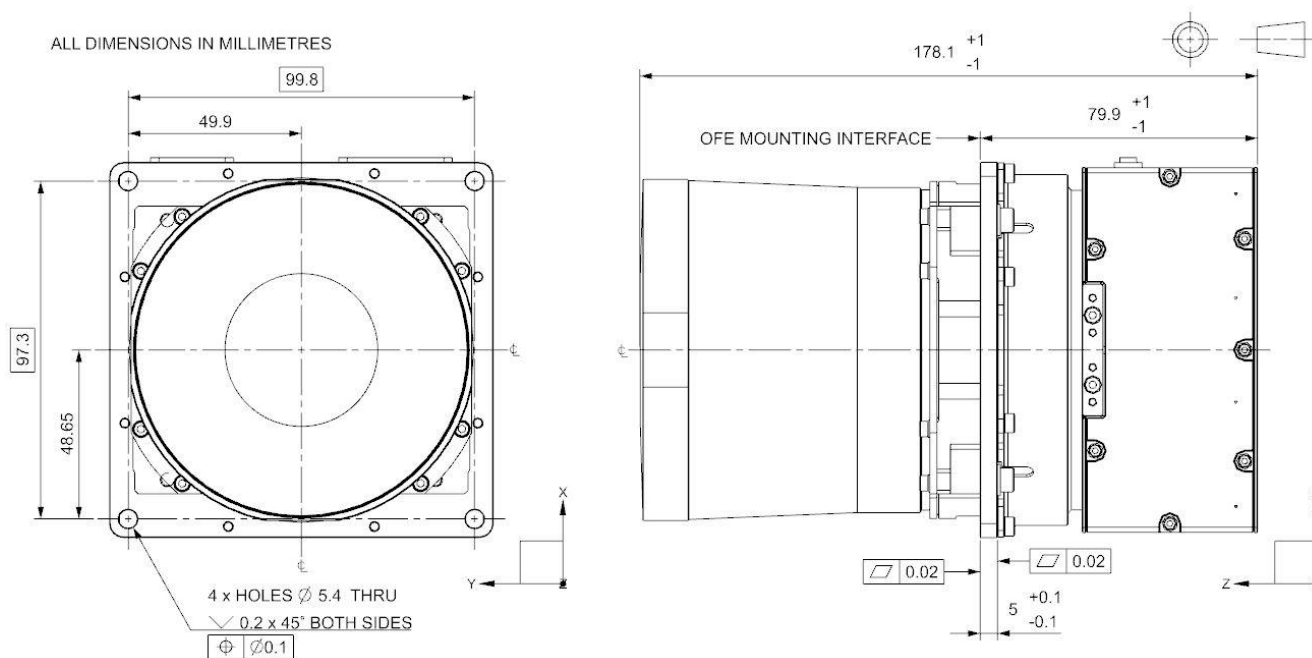


Figure 4-5: OFE Mounting Interface Dimensions

The details of the OFE Mounting Interface are specified in Table 4-3 below.

Table 4-3: OFE Mounting Interface Specifications

Description	Value
Interface Material	Titanium Grade 5 (Ti-6Al-4V)
Interface Surface Finish	Uncoated Ti-6Al-4V N7 Surface Finish
Flatness	< 20 μm
Number of Mounting Locations	4
Mounting Specification	$\varnothing$ 5.4 mm Through Holes (4 off)
Mounting Interface Plate Thickness	5 mm $\pm$ 0.1 mm
Fastener Torque (for M5 stainless steel, A4-70 fastener). All fasteners shall be staked using Scotch Weld EC2216 adhesive or equivalent.	5 Nm

4.3.2 Interface 4: xScale100 Sensor Unit (with Enclosure) to Satellite Components

The Sensor Unit interfaces mechanically with the satellite components through two thermal control surfaces, which can be used to dissipate heat from the sensor's heat sink to the satellite components via conduction. Refer to Section 4.4.2 for the thermal interface description. The two thermal control surfaces are shown highlighted in blue in Figure 4-6 below.

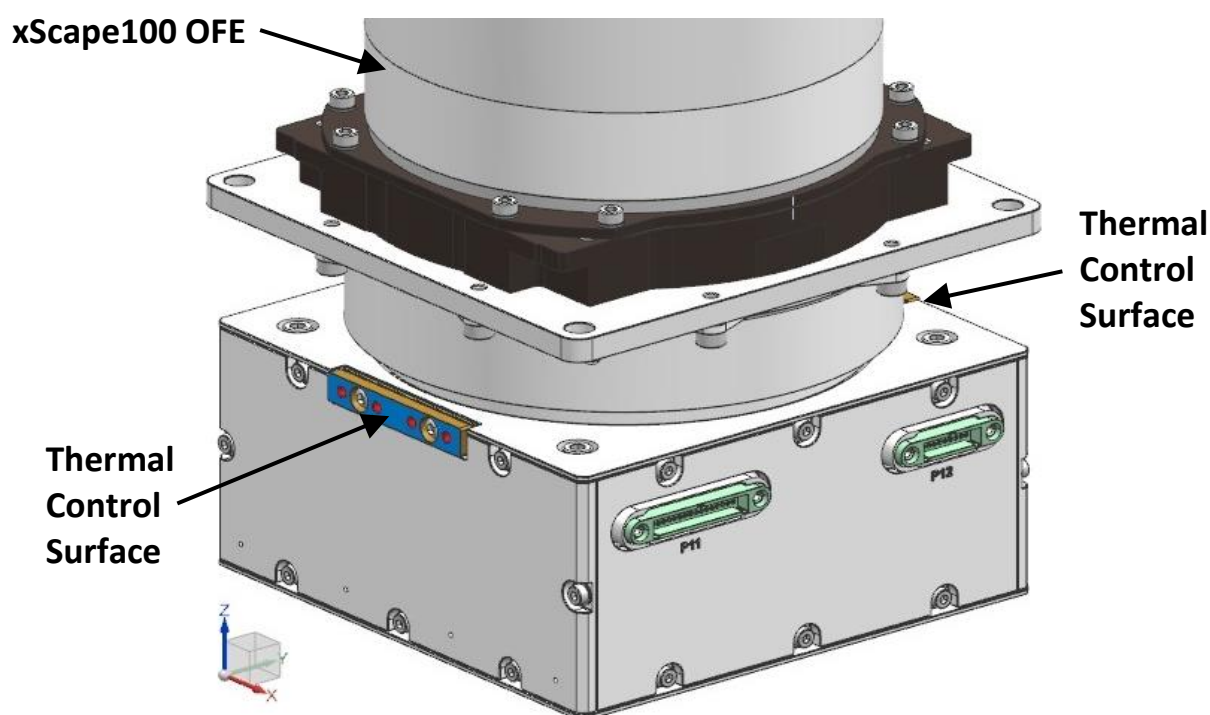


Figure 4-6: Thermal Control Surfaces on Sensor Unit

Figure 4-7 and Figure 4-8 indicate the location of the Sensor Unit's thermal control surfaces relative to the OFE Mounting Interface.

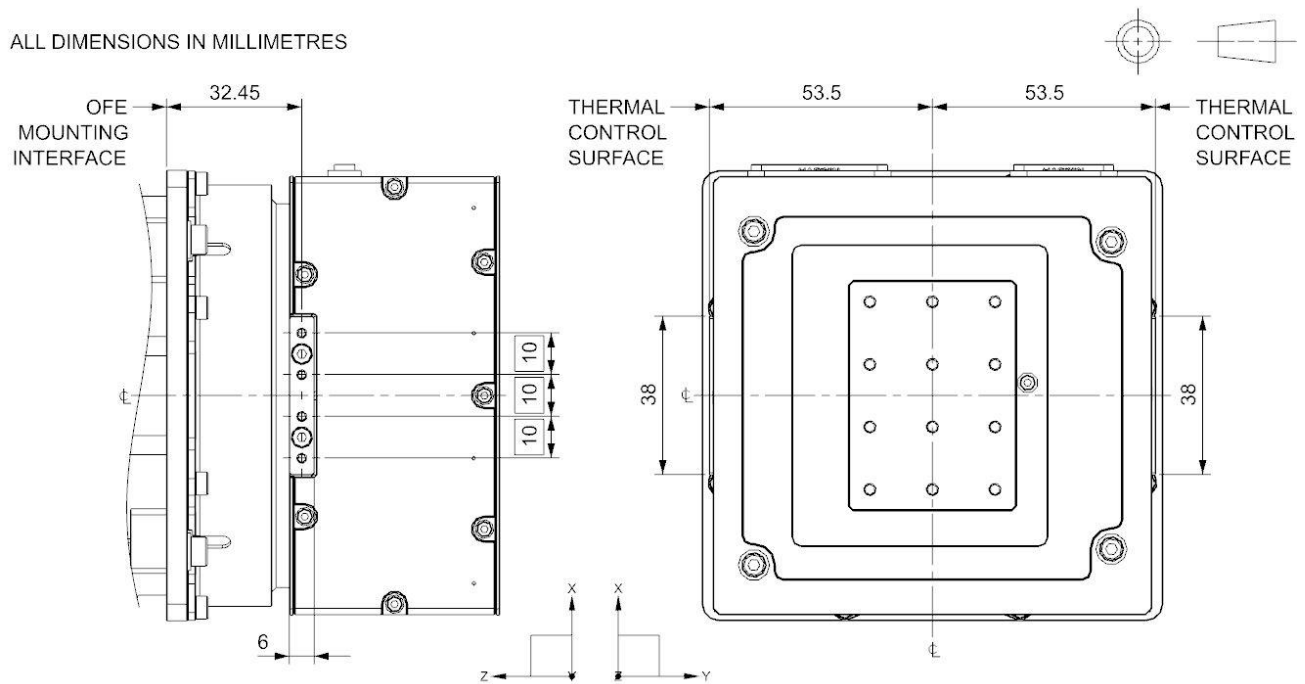


Figure 4-7: Location of Sensor Unit Thermal Control Surfaces [1]

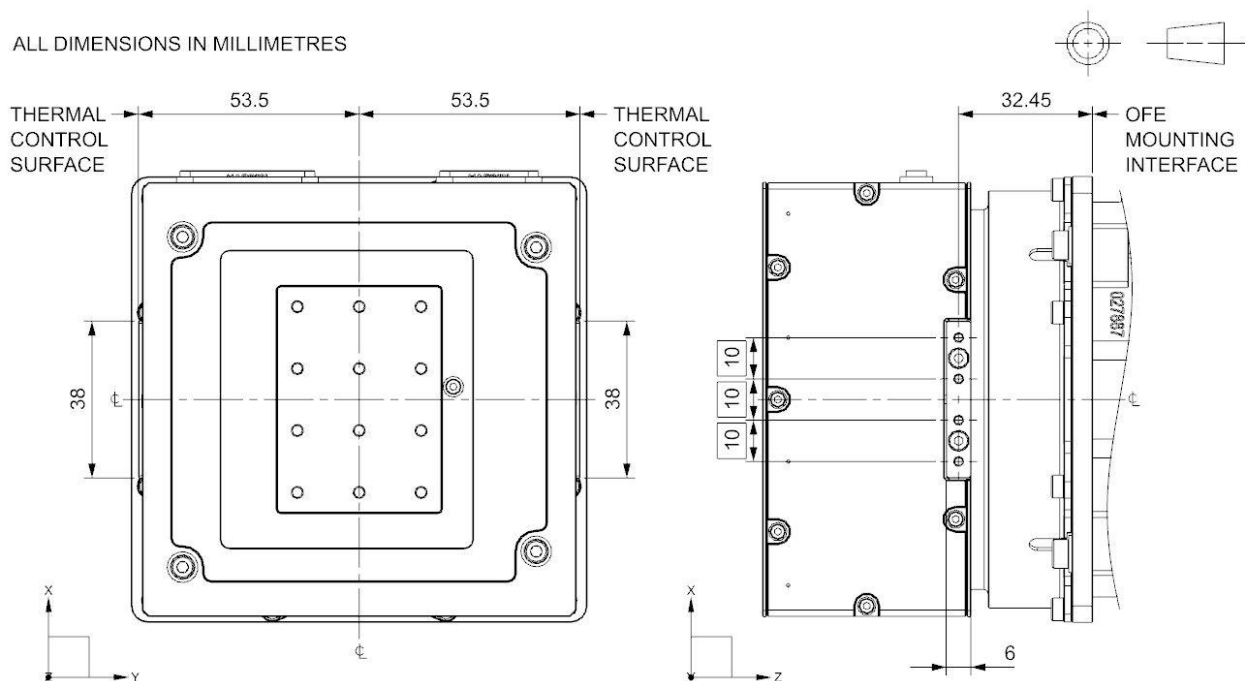


Figure 4-8: Location of Sensor Unit Thermal Control Surfaces [2]

Figure 4-9 shows the bolt pattern provided in the each of the Sensor Unit's thermal control surfaces, as well as the relevant thread specifications. This bolt pattern shall be used if heat straps need to be mounted to these thermal control surfaces. When mounting heat straps to these thermal control surfaces, it shall be ensured that there is sufficient compliance in the heat straps, so as to not induce any strain on the Sensor Unit during integration of the heat straps or during on-orbit operation of the Imager.

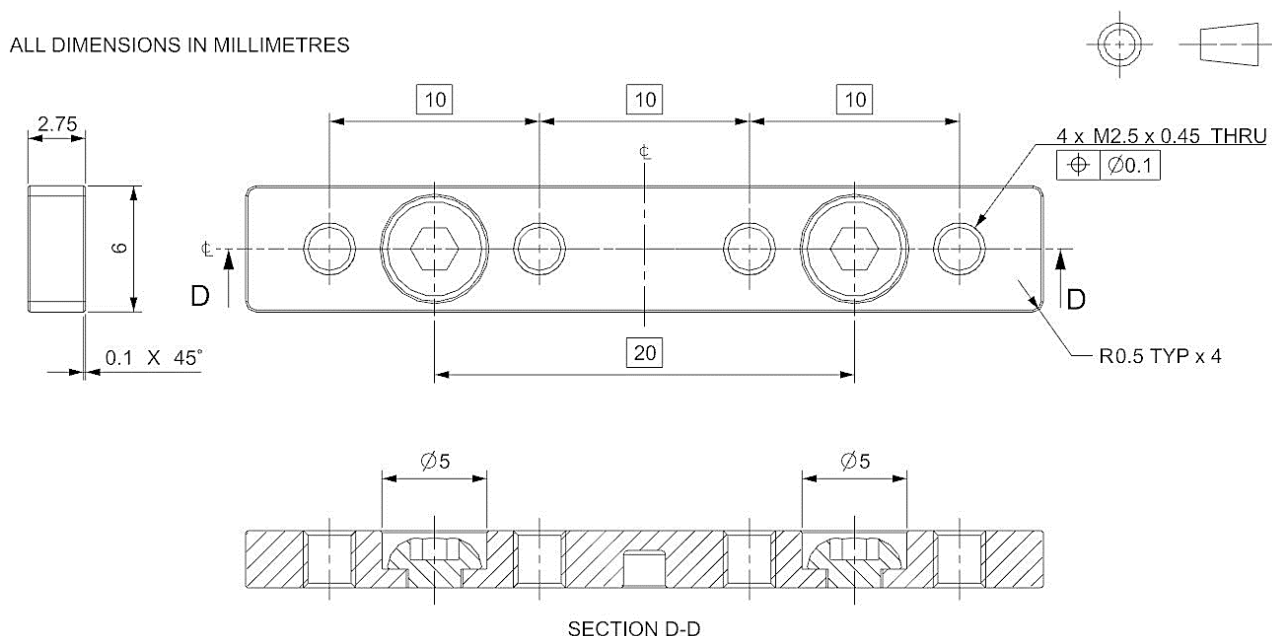


Figure 4-9: Mounting Details on Each Sensor Unit Thermal Control Surface

The specifications of the Sensor Unit thermal control surfaces are provided in Table 4-4 below.

Table 4-4: Specifications of Thermal Control Surface

Description	Value
Interface Material	Copper C11000
Interface Surface Specification	Uncoated Copper, N7 surface finish
Mounting Details (per interface)	4 x M2.5 mm Threaded Holes (see Figure 4-9)
Thread Depth	2.75 mm
Interface Cross Sectional Area (per interface)	$157 \times 10^{-6} \text{ m}^2$

4.4 Thermal Interfaces

4.4.1 Interface 3: xScape100 OFE to Satellite Components

The mounting interface between the OFE and applicable satellite components facilitates heat transfer through thermal conduction between the OFE and the satellite components.

The Mounting Interface Temperature (MIT) is defined as the *temperature of the mechanical mounting interface on the OFE Mounting Interface Plate* (not on the satellite bus structure). That is, the MIT refers to the temperature of the OFE Mounting Interface (as defined in Figure 4-5 and highlighted in blue in Figure 4-10) and NOT to the temperature of the satellite bus structure.

Figure 4-10 below shows the modelled OFE Mounting Interface (highlighted in blue) on the OFE Mounting Interface Plate. The MIT is the temperature of this interface. The MIT is to be considered as a boundary condition to the OFE. Thermal models should thus include the thermal resistance of the bolted interface between the OFE Mounting Interface and the satellite structure. For further details regarding the thermal modelling, refer to [8].

The athermal design of the Imager has been optimised for a MIT range of 5 °C to 20 °C. Therefore, during imaging, the MIT shall be maintained between 5 °C and 20 °C, to achieve optimal optical performance.

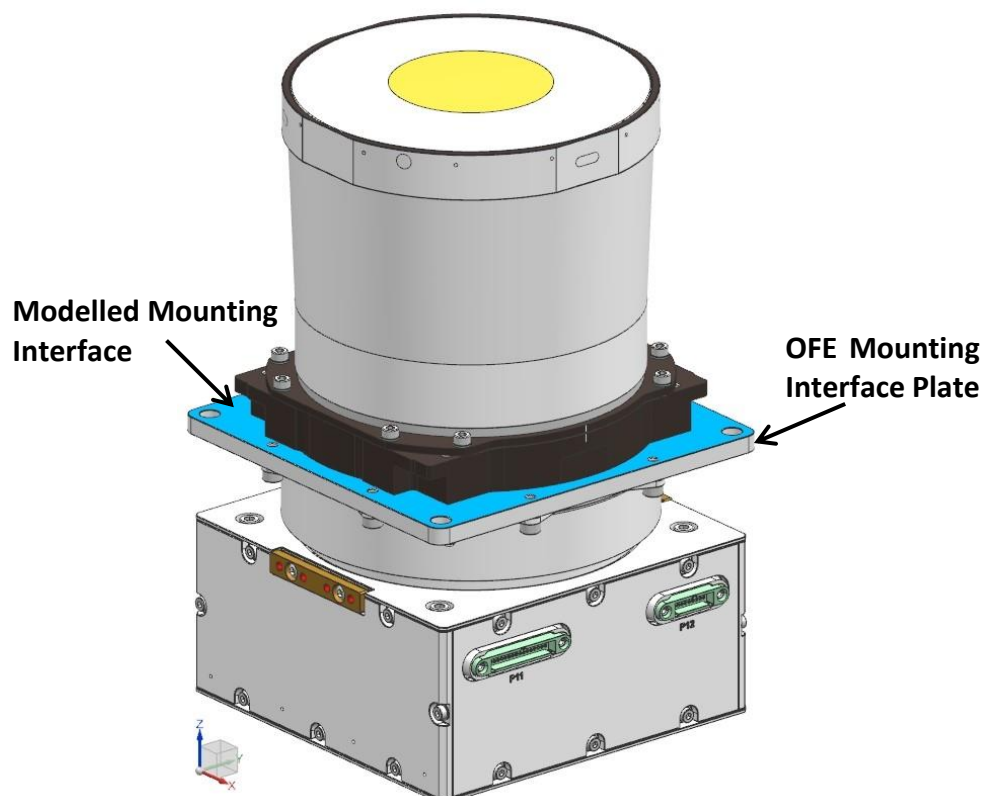


Figure 4-10: Modelled Mounting Interface on OFE Mounting Interface Plate

In addition, thermal energy is also exchanged (by means of radiation) between the OFE and the surrounding satellite components. The outer surface of the OFE's Main Body Upper component is covered with an aluminium thermal film (shown highlighted in blue in Figure 4-11 below). This thermal film has a low thermal emissivity and reduces the radiation exchange with the internal surfaces of the satellite structure. This in turn reduces the on-orbit temperature fluctuation of the OFE's first optical element (Entrance Lens).

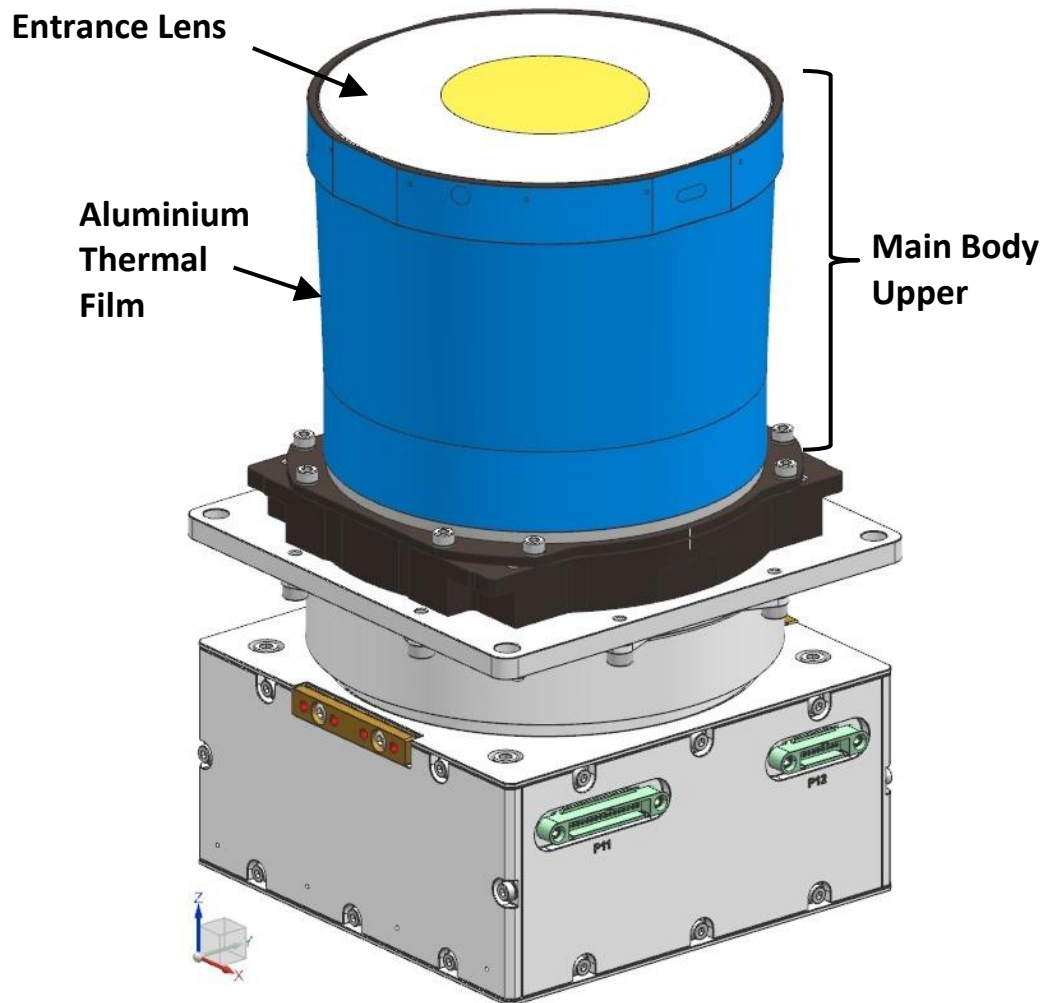


Figure 4-11: Thermal Film on Main Body Upper

4.4.2 Interface 5: xScape100 Sensor Unit (with Enclosure) to Satellite Components

Two thermal interfaces are provided to facilitate heat transfer via conduction between the Sensor Unit and the applicable satellite components (refer to Section 4.3.2 for the mechanical interface description). These two interfaces are connected (internally) via heat straps to a heat sink, which in turn is directly mounted to the sensor (see Figure 3-3). These two interfaces act as thermal control surfaces for the sensor and can thus

be used to control the temperature of the sensor. These thermal control surfaces are conductively insulated from the rest of the xScape100 Electronics Enclosure.

During imaging, the temperature of these thermal control surfaces on the Sensor Unit shall not exceed 25 °C. If the thermal analysis shows that the temperatures of these surfaces exceed 25 °C, before the required imaging time is reached, heat straps shall be mounted to these two thermal control surfaces to control their temperatures (and therefore the temperature of the sensor).

4.4.3 Interface 6: xScape100 Control Electronics to Satellite Components

The surface on the bottom of the Control Electronics (CE) Cover Plate is shown highlighted in blue in Figure 4-12 below. This surface acts as the thermal control surface for the CE.

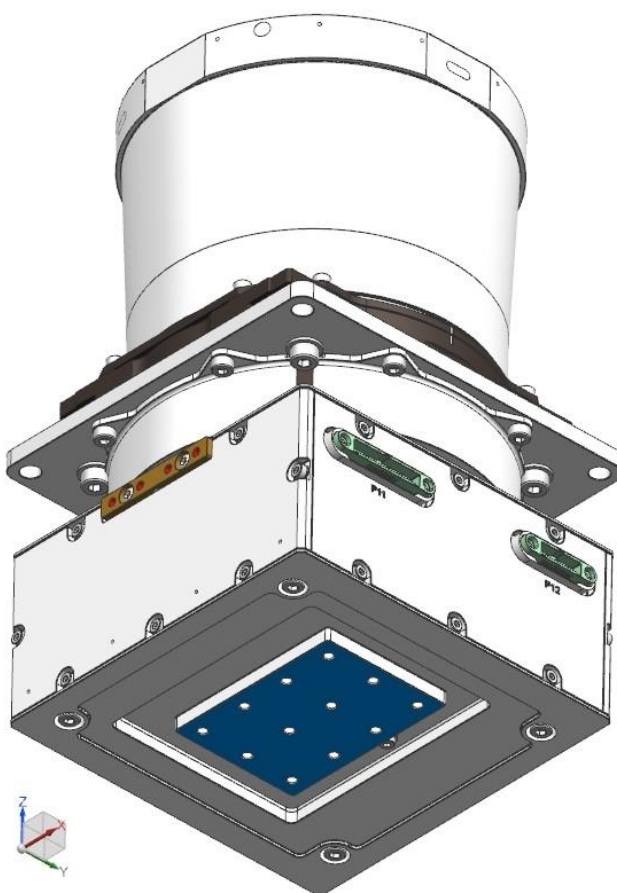


Figure 4-12: Thermal Control Surface on CE Cover Plate

During **imaging** the temperature of the highlighted surface in Figure 4-12 (the thermal control surface) shall be maintained between 5 °C and 30 °C. In addition, during **operation (excluding imaging)** the temperature of the thermal control surface shall be maintained within the operating temperature range (-10 °C to 50 °C) of the Imager, as specified in Section 5.4.2.

If the thermal analysis shows that the temperature of this thermal control surface on the CE Cover Plate falls outside the temperature range of 5 °C to 30 °C (during imaging) OR outside the temperature range of -10 °C to 50 °C (during operation and not imaging), heat straps shall be mounted to the thermal control surface to control the CE Cover Plate temperature. NOTE: The temperature of the CE PCB and the temperatures of the electronic components on the CE PCB may exceed the maximum temperature of the thermal control surface on the CE Cover Plate.

Figure 4-13 shows the bolt pattern provided in the Cover Plate's thermal control surface, as well as the relevant thread specifications. This bolt pattern shall be used if heat straps need to be mounted to the CE Cover Plate.

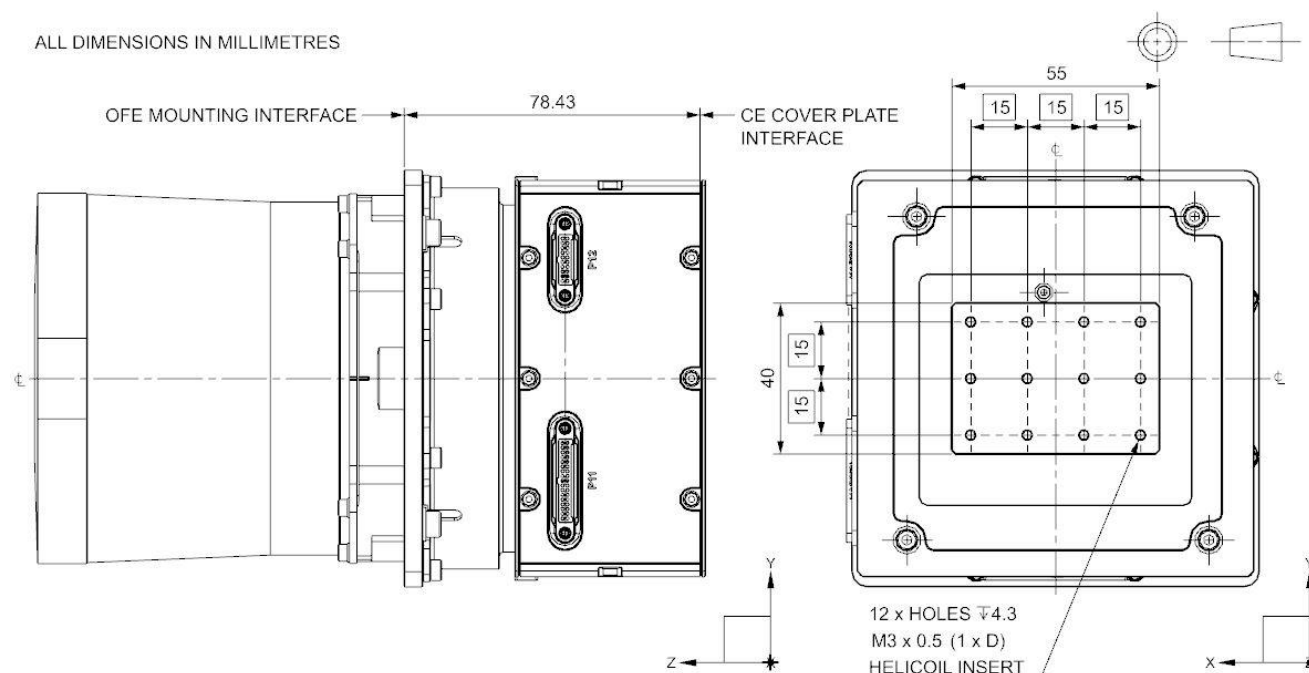


Figure 4-13: Mounting Details on CE Cover Plate Thermal Control Surface

The specifications of the CE Cover Plate Thermal Control Surface are provided in Table 4-5 below.

Table 4-5: Specifications of CE Cover Plate Thermal Control Surface

Description	Value
Interface Material	Aluminium 6082-T6
Interface Surface Specification	N7 Surface Finish Alodined Clear (MIL-DTL-5541 Type II)
Surface Flatness	< 0.05 mm
Fixation Details	12 x M3 mm threaded holes (see Figure 4-13)
Thread Depth	3 mm
Interface Cross Sectional Area	$2.08 \times 10^{-3} \text{ m}^2$

In addition, thermal energy is also exchanged (by means of radiation) between the CE and the surrounding satellite components. No specific requirement is placed on the amount of energy transmitted via radiation

4.4.4 Interface 7: xScape100 Electronics Enclosure to Satellite Components

Thermal energy is also exchanged (by means of radiation) between the xScape100 Electronics Enclosure and the surrounding satellite components. No specific requirement is placed on the amount of energy transmitted via radiation.

4.5 Electrical Interfaces

4.5.1 Interface 8: Satellite Components to xScape100 Control Electronics

This interface consists of two electrical connectors, P11 and P12, that act as interface between the CE and the applicable satellite component(s). The function of this interface is to transmit power, control, and data to and from the CE. The connectors are located on the side panel (facing in the +x-direction) of the xScape100 Electronics Enclosure. Refer to Figure 4-14 below for the connector locations.

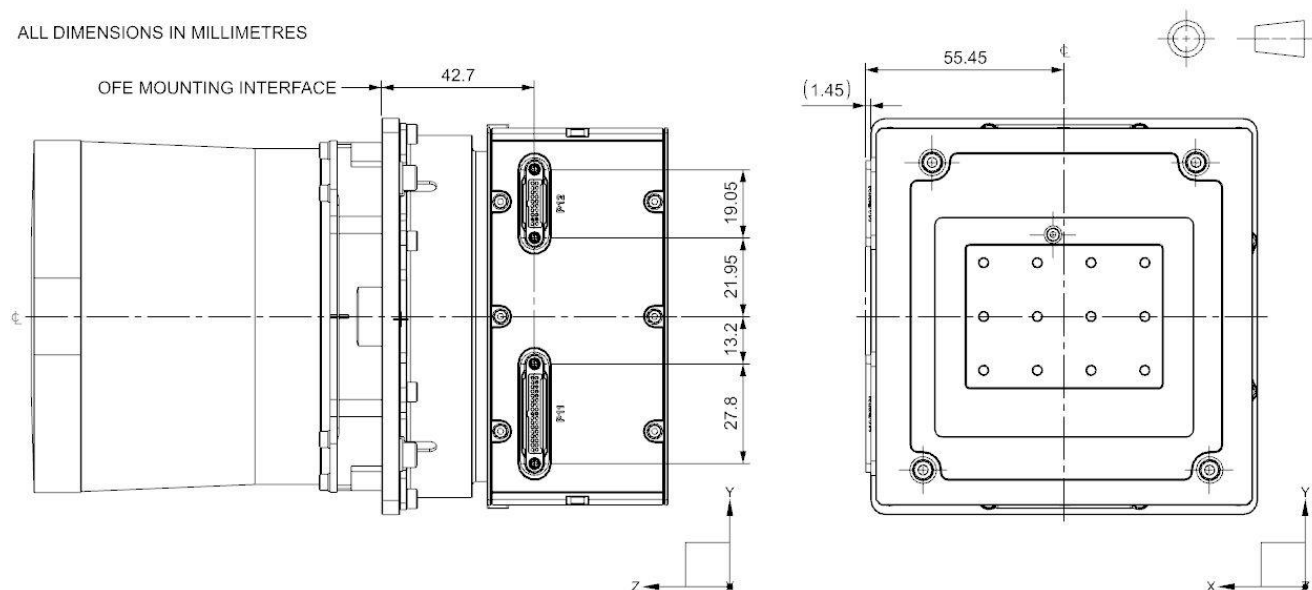


Figure 4-14: P11 and P12 Connector Locations

4.5.1.1 Physical Electrical Interfaces

Figure 4-15 shows a rendering of the enclosure “+x” side panel including the physical connector pin locations.

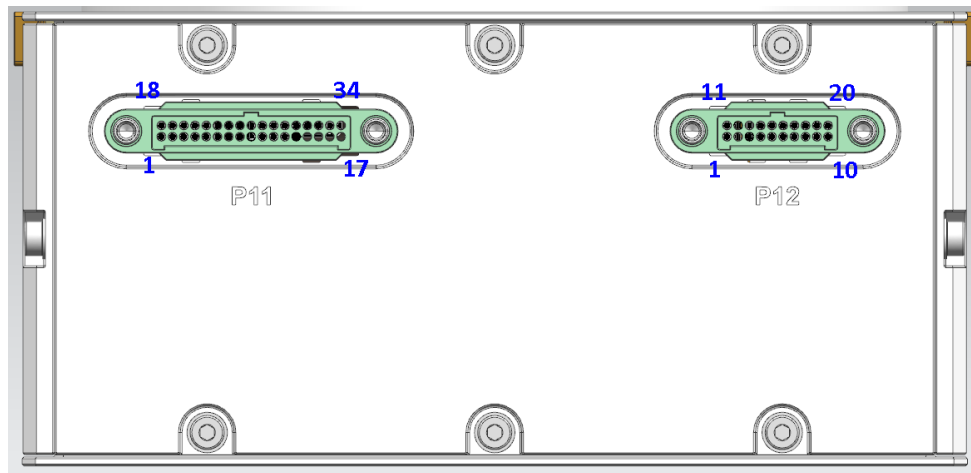


Figure 4-15: Electrical Connector Pin Locations

4.5.1.1.1 P11 Interface

The P11 physical electrical interface provides access to an SPI control interface as well as LVDS, USART and SpaceWire data interfaces. It also includes a PPS input supporting LVDS signalling levels.

P11 Interface Connector

A Harwin G125-3243496M2 connector is used. A rendering of the connector is shown in Figure 4-16. This connector mates to Harwin G125-2243496F1.



Figure 4-16: P11 Interface Connector (Harwin G125-3243496M2)

P11 Interface Pinout

The following table lists the P11 Interface connector pinouts.

Table 4-6: P11 Interface Connector Pinout

Pin	Signal	Pin	Signal
1	SpW Inner Shield ¹	18	SpW Inner Shield ¹
2	SpW SOUT-	19	SpW DOUT-
3	SpW SOUT+	20	SpW DOUT+
4	SpW Shield ²	21	SpW Shield ²
5	SpW SIN-	22	SpW DIN-
6	SpW SIN+	23	SpW DIN+
7	HS_Clk Shield ³	24	HS_D0 Shield ³
8	HS_Clk- / US_Clk-	25	HS_D0- / US_TxData-
9	HS_Clk+ / US_Clk+	26	HS_D0+ / US_TxData+
10	HS_RR Shield ³	27	HS_Ctrl Shield ³
11	HS_RR- / US_nCTS-	28	HS_Ctrl- / US_EOF-
12	HS_RR+ / US_nCTS+	29	HS_Ctrl+ / US_EOF+
13	SPI_SCK Shield ⁴	30	SPI_MISO Shield ⁵
14	SPI_SCK	31	SPI_MISO
15	SPI_nSEL	32	SPI_MOSI
16	Not Used	33	LVDS_PPS Shield ³
17	LVDS_PPS-	34	LVDS_PPS+

1. SpaceWire Inner shield for Type AL (legacy) links only.
2. SpaceWire Cable & inner shield for both Type A and Type AL (legacy) links.
3. Use as shield for corresponding LVDS pair for improved signal integrity.
4. Connect and couple or twist with SPI_SCK for better signal integrity.
5. Connect and couple or twist with SPI_MISO for better signal integrity.

4.5.1.1.2 P12 Interface

The P12 physical electrical interface provides access to the Power interface as well as both I²C and RS-422 control interfaces. It also includes a PPS input supporting LVCMOS signalling levels.

P12 Interface Connector

A Harwin G125-3243496M2 connector is used. A rendering of the connector is shown in Figure 4-16. This connector mates to Harwin G125-2242096F1.



Figure 4-17: P12 Interface Connector (Harwin G125-3242096M2)

P12 Interface Pinout

The following table lists the P12 Interface connector pinouts.

Table 4-7: P12 Interface Connector Pinout

Pin	Signal	Pin	Signal
1	Power Control	11	LVCMOS_PPS
2	Power Control Return	12	LVCMOS_PPS Return
3	I ² C SCL	13	I ² C SDA
4	I ² C SCL Shield ¹	14	I ² C SDA Shield ²
5	RS-422_RX+	15	RS-422_TX+
6	RX-422_RX-	16	RS-422_TX-
7	RS-422_RX Shield ³	17	RS-422_TX Shield ³
8	Power 5V	18	Power Return
9	Power 5V	19	Power Return
10	Power 5V	20	Power Return

1. Connect and couple or twist with I²C_SCL for improved signal integrity.
2. Connect and couple or twist with I²C_SDA for improved signal integrity.
3. Use as shield for corresponding RS-422 pair for improved signal integrity.

4.5.1.2 Power Interface

The Imager requires a regulated 5 V $\pm$ 5% Direct Current (DC) supply at a current of at least 1.8 A. This DC input power supply is filtered on the CE using a discrete Electro-Magnetic Interference (EMI) filter. A simplified schematic of the Power Interface is shown in Figure 4-18.

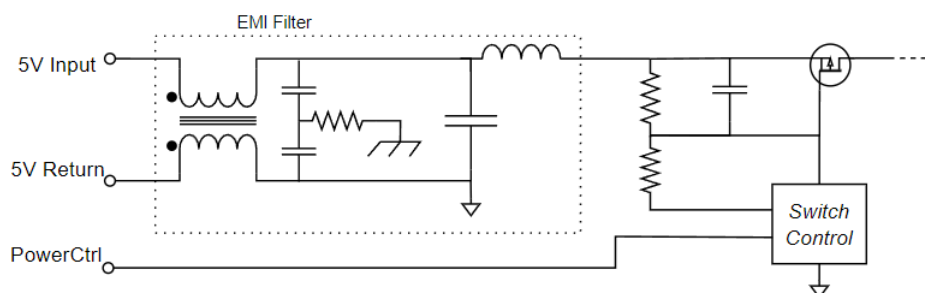


Figure 4-18: Imager Power Interface Schematic

The standard product nominal DC current consumption for the different modes in the operational state is given in Table 4-8.

Table 4-8: Nominal Standard Imager Power Consumption at 25 °C

Operational Mode	Condition	MonoScape100 / TriScape100	MultiScape100	HyperScape100
Idle Mode ⁽¹⁾	Beginning of Life	550 mA, 2.75 W	650 mA, 3.25 W	800 mA, 4.00 W
	After 25 krad TID	580 mA, 2.90 W	690 mA, 3.45 W	850 mA, 4.25 W
Sensor Enabled ⁽²⁾	Beginning of Life	1100 mA, 5.50 W	1200 mA, 6.00 W	1350 mA, 6.75 W
	After 25 krad TID	1160 mA, 5.80 W	1270 mA, 6.35 W	1430 mA, 7.15 W
Imaging Mode ⁽³⁾	Beginning of Life	1350 mA, 6.75 W	1400 mA, 7.00 W	1550 mA, 7.75 W
	After 25 krad TID	1430 mA, 7.15 W	1480 mA, 7.40 W	1640 mA, 8.20 W
Readout Mode ⁽⁴⁾	Beginning of Life	550 mA, 2.75 W	650 mA, 3.25 W	800 mA, 4.00 W
	After 25 krad TID	580 mA, 2.90 W	690 mA, 3.45 W	850 mA, 4.25 W

(1) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are static.

(2) CE and FEE are powered on, but no image is being captured.

(3) CE and FEE are powered on, and an image (burst mode for snapshot imagers) is being captured.

(4) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are active.

The nominal DC current consumption of Imagers including image compression option for the different modes in the operational state is given in Table 4-9.

Table 4-9: Nominal Imager Power Consumption at 25 °C Including Image Compression

Operational Mode	Condition	MonoScape100 / TriScape100	MultiScape100	HyperScape100
Idle Mode ⁽¹⁾	Beginning of Life	650 mA, 3.25 W	750 mA, 3.75 W	900 mA, 4.50 W
	After 25 krad TID	690 mA, 3.45 W	800 mA, 4.00 W	950 mA, 4.75 W
Sensor Enabled ⁽²⁾	Beginning of Life	1200 mA, 6.00 W	1300 mA, 6.50 W	1450 mA, 7.25 W
	After 25 krad TID	1270 mA, 6.35 W	1375 mA, 6.88 W	1540 mA, 7.70 W
Imaging Mode ⁽³⁾	Beginning of Life	1450 mA, 7.25 W	1500 mA, 7.50 W	1650 mA, 8.25 W
	After 25 krad TID	1540 mA, 7.70 W	1590 mA, 7.95 W	1750 mA, 8.75 W
Readout Mode ⁽⁴⁾	Beginning of Life	650 mA, 3.25 W	750 mA, 3.75 W	900 mA, 4.50 W
	After 25 krad TID	690 mA, 3.45 W	800 mA, 4.00 W	950 mA, 4.75 W

(1) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are static.

(2) CE and FEE are powered on, but no image is being captured.

(3) CE and FEE are powered on, and an image (burst mode for snapshot imagers) is being captured.

(4) CE is powered on, but the FEE is off. Control and High-Speed Data interfaces are active.

The 5 V supply must be supplied by one, or a combination, of the CSK interface connector pins H1.47, H1.49, H1.51, H2.13, H2.15, H2.16, H2.25, and H2.26, or the high-speed interface connector pins P5.95, P5.96, P5.97, P5.98, P5.99 and P5.100. The ground connection is made by one, or a combination, of CSK interface connector pins H2.29, H2.30, and H2.32, or the high-speed interface connector pins P5.86, P5.89, P5.90, P5.91, P5.93, P5.94.

The CE has an optional power switch onboard that can be used if the 5 V supplied to the CE is not switched. By default, the CE operates in “Bus Switch” power mode and the CE onboard power switch is disabled. If the “User-Controlled Switch” power mode is selected in the product configuration sheet, the power switch control line ‘PowerCtrl’ can be routed to one of the CSK interface connector pins H1.4, H1.5, H1.6, H1.7, H1.8, H1.9, H1.10, H1.11, H1.13, H1.14, H1.15, H1.16, H2.17, H2.18, H2.19 or H2.20, or the high speed connector pin P5.87. The ‘PowerCtrl’ control signal is active high. To turn on the CE, the ‘PowerCtrl’ control signal must be driven high by applying a voltage between 2.5 V and 5.0 V for at least 100 ms. The Imager can be turned off by driving ‘PowerCtrl’ low to between 0 V and 0.5 V for at least 100 ms.

In both power modes, the CE will power-cycle itself in the event of a radiation induced overcurrent condition. The different power switching possibilities are shown in Table 4-10.

Table 4-10: Power Switching Alternatives

Power Mode	Description	Over-current
Bus Switched *	External 5 V supply is switched. CE power switch is disabled. Power Control line is not used.	Power-cycle
User-Controlled Switch	External 5 V supply stays ON. Switch ON while Power Control line is driven high. Switch OFF while Power Control line is driven low.	Power-cycle

* Default

4.5.1.3 Control and Data Interface

Details of the standard control and data interfaces can be found in [2].

Details of the optional SpaceWire interface can be found in [3].

Details of the optional RS422/RS485 control interface can be found in [4].

Details of the optional USART data interface can be found in [5].

Details of the optional data encryption interface can be found in [6].

Details of the optional image compression interface can be found in [7].

4.5.1.3.1 I²C Control Interface

The Imager provides a single I²C bus connection as specified in [2]. The I²C control interface is always available on the P12 interface.

Table 4-11: I²C on P12 Interface Pinout

Signal	P12
SCL	3
I ² C GND ¹	4
SDA	13
I ² C GND ²	14

1. Connect and couple or twist with SCL for improved signal integrity.
2. Connect and couple or twist with SDA for improved signal integrity.

4.5.1.3.2 SPI Control Interface

The Imager provides an SPI interface as specified in [2]. The SPI interface is optional and may be enabled on the P11 interface as shown in Table 4-12.

Table 4-12: SPI on P11 Interface Pinout

Signal	P11
SCK	14
SPI GND ¹	13
nSEL	15
MOSI	32
MISO	31
SPI GND ²	30

1. Connect and couple or twist with SCK for improved signal integrity.
2. Connect and couple or twist with MISO for improved signal integrity.

4.5.1.3.3 RS-422 Control Interface

The Imager provides an RS-422 interface as specified in [4]. The RS-422 control interface may be enabled on the P12 interface as shown in Table 4-13. Termination resistors may be fitted to RX and/or TX, depending on the chosen configuration. The RS-422 control interface supports various protocols; see [4] for details.

Table 4-13: RS-422 on P12 Interface Pinout

Signal	P12
TX+	15
TX-	16
TX Shield	17
RX+	5
RX-	6
RX Shield	7

4.5.1.3.4 SpaceWire Control and/or Data Interface

The Imager provides a SpaceWire interface as specified in [3]. A single SpaceWire interface may provide a control, data or both control and data interfaces. The SpaceWire interface is optional and may be enabled on the P11 interface as shown in Table 4-14.

Table 4-14: SpaceWire on P11 Interface Pinout

Signal	P11
Shield ¹	21
DIN+	23
DIN-	22
Shield ¹	4
SIN+	6
SIN-	5
DOUT Inner Shield ²	18
DOUT+	20
DOUT-	19
SOUT Inner Shield ²	1
SOUT+	3
SOUT-	2

1. Cable & inner shield for both Type A and Type AL (legacy) links.
2. Inner shield for Type AL (legacy) links only.

4.5.1.3.5 LVDS Data Interface

The Imager provides an LVDS data interface as specified in [2]. A single lane LVDS interface is enabled on the P11 interface as shown in Table 4-15. Note that the LVDS and USART data interfaces cannot be enabled at the same time. Refer to 4.5.1.3.6.

Table 4-15: LVDS on P11 Interface Pinout

Signal	P11
Clk Shield	7
Clk+	9
Clk-	8
Ctrl Shield	27
Ctrl+	29
Ctrl-	28
D0 Shield	24
D0+	26
D0-	25
RR Shield	10
RR+	12
RR-	11

4.5.1.3.6 USART Data Interface

The Imager provides an USART data interface as specified in [5]. The USART interface is optional and may be routed to the P11 interface as shown in Table 4-16. Note that the LVDS and USART data interfaces cannot be enabled on the same connector at the same time. Refer to 4.5.1.3.5.

Table 4-16: USART on P11 Interface Pinout

Signal	P11
Clk Shield	7
Clk+	9
Clk-	8
nCTS Shield	10
nCTS+	12
nCTS-	11
TxDa Shield	24
TxDa+	26
TxDa-	25
EOFS Shield	27
EOFS+	29
EOFS-	28

4.5.1.3.7 PPS Input

The Imager provides both LVCMOS and LVDS PPS input interfaces that can be used to create accurate timestamp metadata within the image data. The PPS input interface is optional and may be routed to the P11 interface or the P12 interface as shown in Table 4-17.

Table 4-17: PPS on High-Speed Interface Pinout

Signal	P11	P12
LVCMOS_PPS_IN		11
LVCMOS_PPS_GND		12
LVDS_PPS Shield	33	
LVDS_PPS+	34	
LVDS_PPS-	17	

4.5.1.4 Grounding

The enclosure connected to the CE Signal GND through multiple broadband 100 nF ceramic capacitors to facilitate grounding.

4.6 Environmental Interfaces

4.6.1 Thermal Radiation Interface

4.6.1.1 Interface 9: xScape100 OFE to Environment

The front aperture of the OFE exchanges thermal energy (by means of radiation) with the space environment. A thermal coating has been applied, through a vapour deposition process, to the central area of the first optical element (Entrance Lens) of the OFE. This coating reduces radiation exchange between the Entrance Lens and the environment, resulting in an increased average temperature of the Entrance Lens during orbit. The position of the thermal coating is shown in Figure 4-19 below.

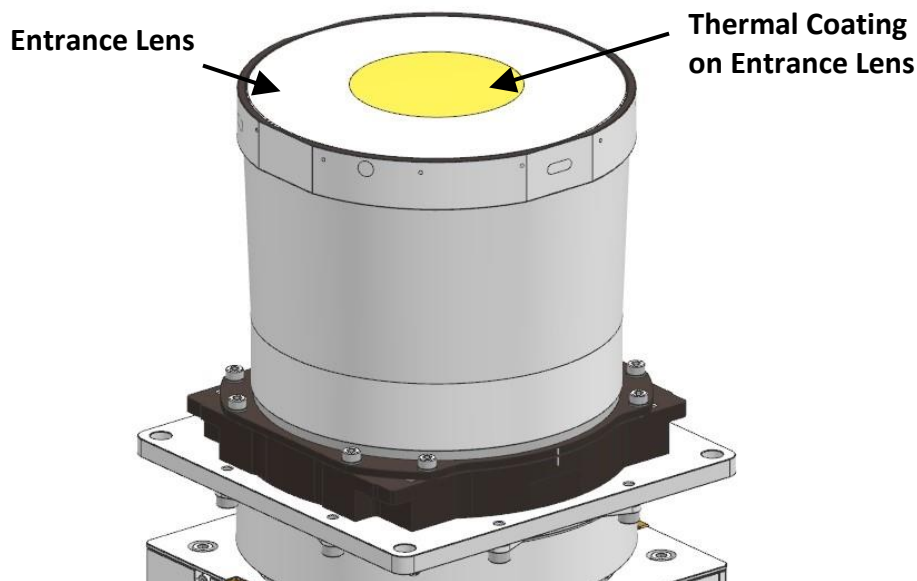


Figure 4-19: Thermal Coating on Entrance Lens

4.6.2 Cosmic Radiation Interfaces

4.6.2.1 Interface 10: Environment to xScape100 (with Enclosure)

There is an interface between the space environment and the xScape100 (with Enclosure) through which cosmic radiation is transmitted to the xScape100 (with Enclosure).

5. Environmental Requirements

5.1 Transportation

5.1.1 Temperature

During transportation and in a non-operating condition the maximum temperature of the Imager assembly shall not exceed 50 °C and the minimum temperature of the Imager assembly shall not be less than -10 °C.

5.1.2 Humidity

The humidity during transportation shall be less than 80%, non-condensing.

5.1.3 Shock/Vibration

All handling operations shall limit the peak acceleration exposure of the Imager to less than 25 g, in any direction.

5.1.4 Cleanliness

During transportation, the Imager shall be kept in an environment with a cleanliness level equal to or better than International Organization for Standardization (ISO) class 8, as per ISO 14644-1:2015 standards.

5.2 Storage

During storage it is assumed that the Imager will not be in motion, therefore vibration and shock loading conditions are not relevant. During storage, the following conditions shall be adhered to:

5.2.1 Temperature

During prolonged storage and in a non-operating condition the maximum temperature of the Imager shall not exceed 30 °C and the minimum temperature of the Imager shall not be less than 10 °C.

5.2.2 Humidity

The humidity during prolonged storage shall be between 30% and 60%, non-condensing.

5.2.3 Cleanliness

The Imager shall be stored in an environment with a cleanliness level equal to or better than ISO class 8, as per ISO 14644-1:2015 standards.

5.3 Assembly, Integration and Testing

All Assembly, Integration and Testing (AIT) procedures shall be performed in an ISO class 8 cleanroom, as per ISO 14644-1:2015 standards, or better. In addition, during AIT the following conditions shall always be adhered to:

5.3.1 Shock

All handling operations shall limit shock exposure of the complete Imager to less than 30 g for 10 ms, in any direction.

5.3.2 Mechanical Interface with OFE

During all AIT procedures, all mechanical mating interfaces with the OFE shall cause zero relative displacement (in the x-y plane) between any of the OFE's four mounting points.

5.4 On-Orbit

All requirements listed in this section shall be adhered to *concurrently* to ensure both survival and performance of the product. This means the requirements detailed in Section 5.4.1 through Section 5.4.5 shall all be adhered to *concurrently*.

5.4.1 Survivable Temperature

To ensure survival, the maximum temperature of the Imager shall not exceed 65 °C and the minimum temperature of the Imager shall not be less than -25 °C.

5.4.2 Operating Temperature

During operation, the maximum temperature of the Imager shall not exceed 50 °C and the minimum temperature of the Imager shall not be less than -10 °C. This operating temperature range is applicable to all Imager components, including the sensor, but excluding the thermal control surfaces on the Sensor Unit (see Section 4.4.2) and on the CE Cover Plate (see Section 4.4.3).

During **imaging**, the temperature of the thermal control surfaces on the Sensor Unit shall not exceed 25 °C. Similarly, during **imaging** the temperature of the thermal control surface on the CE Cover Plate shall be maintained between 5 °C and 30 °C. In addition, during **operation (excluding imaging)** the temperature of the thermal control surface on the CE Cover Plate shall be maintained within the operating temperature range of -10 °C to 50 °C (see Section 4.4.3).

The operating temperature ranges specified above are *not* applicable to the CE PCB and the electronic components on the CE PCB. The temperatures of these components are controlled by controlling the temperature of the thermal control surface on the CE Cover Plate.

5.4.3 Mounting Interface Temperature Requirement

The Mounting Interface Temperature (MIT) is defined as the *temperature of the mechanical mounting interface on the OFE Mounting Interface Plate* (not on the satellite bus structure).

The athermal design of the Imager has been optimised for a MIT range of 5 °C to 20 °C. Therefore, during imaging, the MIT shall be maintained between 5 °C and 20 °C, to achieve optimal optical performance.

5.4.4 Allowable Temperature Gradients

Thermal reference points are defined on the Imager in order to specify the allowable thermal gradients. The top of the Main Body Upper has four flat spots, with each feature being perpendicular to either the x- or y-axis of the Imager. The centres of each of these four flat spots are used as the locations of the four thermal

reference points on the Main Body Upper. Similarly, the four thermal reference points on the Main Body Lower are also located in the centres of each of the four flat spots on the Main Body Lower.

The axial gradient is defined as the difference in temperature between the average temperature of the four thermal reference points on the Main Body Upper and the average temperature of the four thermal reference points on the Main Body Lower (as shown in Figure 5-1). During operation, the maximum axial temperature gradient shall not exceed 10 °C.

The transverse thermal gradients, in the x- and y-directions, are defined as the difference in temperature between two opposing reference point temperatures on the Main Body Upper (as shown in Figure 5-1). During operation, the maximum transverse temperature gradient shall not exceed 2 °C. The locations on the Imager between which the temperature gradients are defined are shown in Figure 5-1 below.

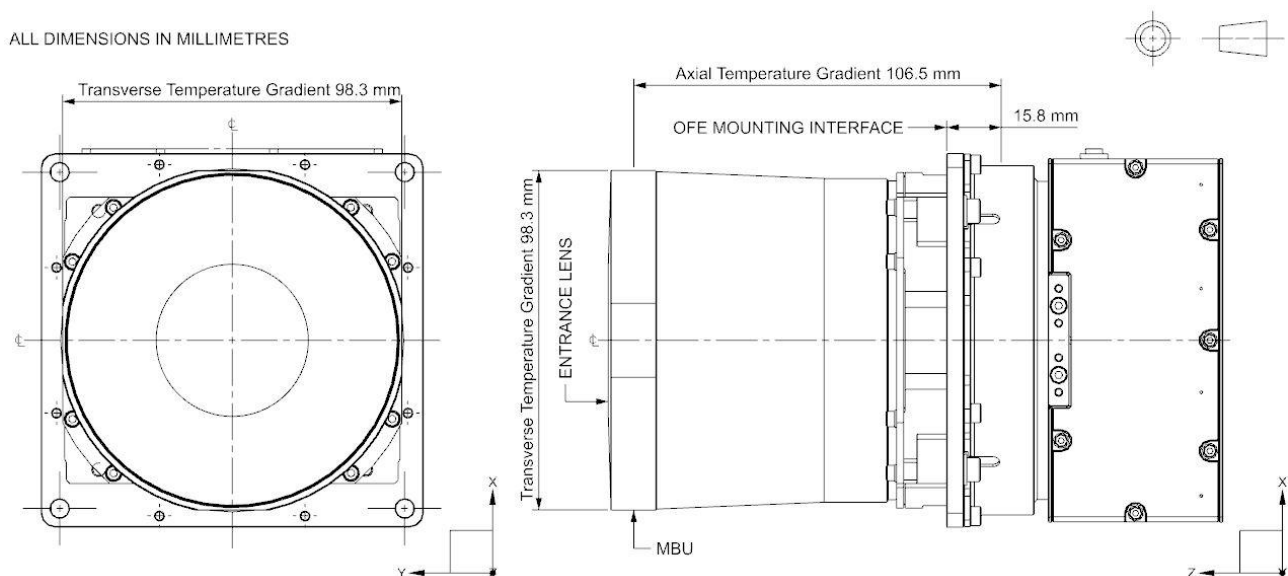


Figure 5-1: Allowable Temperature Gradients

5.4.5 Outgassing of Satellite Components

Material used in satellite components, which are near the Imager, shall have a maximum Total Mass Loss (TML) of less than 1.0% and a maximum Collected Volatile Condensable Material (CVCM) of less than 0.10%. Refer to [1] for a list of material TML and CVCM data.

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